# beijing

Full twelve-station hourly panel. Supplied NA measurements remain missing cells, distinct from absent records.

Completed evidence · 1 sources · 39 retained records

15 evidence-qualified finding(s) across 1 supplied source(s). Required stages completed over eligible records; behavioral search scope is disclosed separately.

**Actions to investigate**

**Window effects meet configured investigation settings**

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t7 · PM2.5

**Window effects meet configured investigation settings**

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t23 · DEWP

**Contiguous equal-value runs merit investigation**

Configured candidate threshold met. Legitimate plateaus, zeros, and operating regimes remain plausible. Review the declarations and bounded witnesses.

investigation_candidate · t29 · RAIN

**Window effects meet configured investigation settings**

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t31 · WSPM

**Contiguous equal-value runs merit investigation**

Configured candidate threshold met. Legitimate plateaus, zeros, and operating regimes remain plausible. Review the declarations and bounded witnesses.

investigation_candidate · t33 · WSPM

**Interpretation boundary**

- No composite score or universal readiness verdict.
- Investigation defaults are not validated defect criteria.
- Supplied evidence does not prove what a production pipeline consumed.

Generated 2026-10-10T22:50:41.807862+00:00 · Schema 1.4 · temporal_reliability_v1

# Declarations

| Evidence / source | Measurement or rule | Retained result |
|---|---|---|
| beijing | Source and base task | {"format":"csv","role":"dataset","task":"dataset_profile","targets":null} |
| beijing | observation | measurement |
| beijing | layout | wide |
| beijing | entity columns | ["station"] |
| beijing | observation keys | [] |
| beijing | clock | {"column":"time","kind":"instant","encoding":"iso","format":null,"epoch_unit":null,"timezone":"Asia/Shanghai","timezone_column":null,"ambiguity":"unresolved","fold_column":n… [JSON] |
| beijing | schedule | {"kind":"fixed","cadence_ns":"3600000000000","anchor":"2013-03-01T00:00:00+08:00","civil_unit":null,"timezone":null,"month_end":"skip","source_id":null,"time_column":null,"e… [JSON] |
| beijing | active intervals | [{"start":"2013-03-01T00:00:00+08:00","end":"2017-03-01T00:00:00+08:00","entity":[],"measurement":null,"closed":"left"}] |
| beijing | PM2.5 | {"name":"PM2.5","column":null,"representation":"quantitative","unit":"ug/m3","meaning":null,"past_observation":false} |
| beijing | PM10 | {"name":"PM10","column":null,"representation":"quantitative","unit":"ug/m3","meaning":null,"past_observation":false} |
| beijing | TEMP | {"name":"TEMP","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false} |
| beijing | PRES | {"name":"PRES","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false} |
| beijing | DEWP | {"name":"DEWP","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false} |
| beijing | RAIN | {"name":"RAIN","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false} |
| beijing | WSPM | {"name":"WSPM","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false} |
| beijing | wd | {"name":"wd","column":null,"representation":"categorical","unit":null,"meaning":null,"past_observation":false} |

16 of 16 metric rows shown. Presentation limits do not change analytical coverage. Long cells are abbreviated; canonical HTML/JSON retains detail.

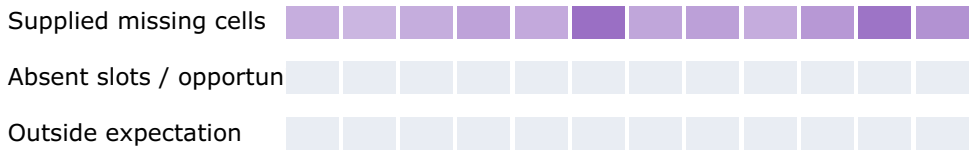
# Chronology

| Evidence / source | Measurement or rule | Retained result |
|---|---|---|
| t1 beijing | rows | 420768 |
| t1 beijing | cells | 3366144 |
| t1 beijing | declared dense missing cells | 0 |
| t1 beijing | parse states / valid | 3366144 |
| t1 beijing | precision counts / second | 3366144 |
| t1 beijing | value states [DEWP] / measurement | DEWP |
| t1 beijing | value states [DEWP] / state | missing |
| t1 beijing | value states [DEWP] / cells | 403 |
| t1 beijing | value states [DEWP] / measurement | DEWP |
| t1 beijing | value states [DEWP] / state | valid |
| t1 beijing | value states [DEWP] / cells | 420365 |
| t1 beijing | value states [PM10] / measurement | PM10 |
| t1 beijing | value states [PM10] / state | missing |
| t1 beijing | value states [PM10] / cells | 6449 |
| t1 beijing | value states [PM10] / measurement | PM10 |
| t1 beijing | value states [PM10] / state | valid |
| t1 beijing | value states [PM10] / cells | 414319 |
| t1 beijing | value states [PM2.5] / measurement | PM2.5 |
| t1 beijing | value states [PM2.5] / state | missing |
| t1 beijing | value states [PM2.5] / cells | 8739 |

20 of 69 metric rows shown. 49 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage. Long cells are abbreviated; canonical HTML/JSON retains detail.

# Continuity and history

Absent observations, supplied missing values and outside-expectation periods have different meanings. Event absence alone does not prove an outage.



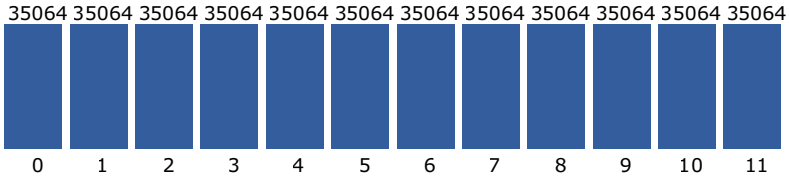

beijing · missing values and absent slots remain separate

| Evidence / source | Measurement or rule | Retained result |
|---|---|---|
| t4 beijing | distinct entities | 12 |
| t4 beijing | distinct series | 96 |
| t4 beijing | history counts / min | 35064 |
| t4 beijing | history counts / max | 35064 |
| t4 beijing | history counts / entity weighted mean | 35064 |
| t4 beijing | history length bins [0] / observations | 4 |
| t4 beijing | history length bins [0] / series | 96 |
| t4 beijing | observed first tick | 1362067200000000000 |
| t4 beijing | observed last tick | 1488294000000000000 |
| t4 beijing | observed entry exit is lifecycle | No |
| t4 beijing | retained histories [DEWP] / entity | ["Aotizhongxin"] |
| t4 beijing | retained histories [DEWP] / measurement | DEWP |
| t4 beijing | retained histories [DEWP] / observations | 35064 |
| t4 beijing | retained histories [DEWP] / cells | 35064 |
| t4 beijing | retained histories [DEWP] / first tick | 1362067200000000000 |
| t4 beijing | retained histories [DEWP] / last tick | 1488294000000000000 |
| t4 beijing | retained histories [DEWP] / span ticks | 126226800000000000 |
| t4 beijing | retained histories [PM10] / entity | ["Aotizhongxin"] |
| t4 beijing | retained histories [PM10] / measurement | PM10 |
| t4 beijing | retained histories [PM10] / observations | 35064 |

20 of 818 metric rows shown. 798 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage. Long cells are abbreviated; canonical HTML/JSON retains detail. Additional charts remain in canonical HTML.

# Behavioral investigation

Thresholds and support settings are configurable investigation defaults. Reference support, sparse windows, zero dispersion, phase exclusions and population mix qualify interpretation.




beijing / PM2.5 · supplied cells per window

| Evidence / source | Measurement or rule | Retained result |
|---|---|---|
| t7 beijing / PM2.5 | Reference | Window 0 · 1362067200000000000 to 1372586100000000000 · unit ug/m3 · support setting 100 |
| t7 beijing / PM2.5 | Window 0 | 35064 cells · missing rate 0.01839493 · invalid 0 · nonfinite 0 · finite 34419 · 12 entities · IQR 92 |
| t7 beijing / PM2.5 | Window 1 | 35064 cells · missing rate 0.01497262 · invalid 0 · nonfinite 0 · finite 34539 · 12 entities · IQR 77 |
| t7 beijing / PM2.5 | Window 2 | 35064 cells · missing rate 0.01662674 · invalid 0 · nonfinite 0 · finite 34481 · 12 entities · IQR 125 |
| t7 beijing / PM2.5 | Window 3 | 35064 cells · missing rate 0.02184577 · invalid 0 · nonfinite 0 · finite 34298 · 12 entities · IQR 79 |
| t7 beijing / PM2.5 | Window 4 | 35064 cells · missing rate 0.02181725 · invalid 0 · nonfinite 0 · finite 34299 · 12 entities · IQR 89 |
| t7 beijing / PM2.5 | Window 5 | 35064 cells · missing rate 0.02760666 · invalid 0 · nonfinite 0 · finite 34096 · 12 entities · IQR 117 |
| t7 beijing / PM2.5 | Window 6 | 35064 cells · missing rate 0.02178873 · invalid 0 · nonfinite 0 · finite 34300 · 12 entities · IQR 75 |
| t7 beijing / PM2.5 | Window 7 | 35064 cells · missing rate 0.02347137 · invalid 0 · nonfinite 0 · finite 34241 · 12 entities · IQR 59 |
| t7 beijing / PM2.5 | Window 8 | 35064 cells · missing rate 0.01514374 · invalid 0 · nonfinite 0 · finite 34533 · 12 entities · IQR 120 |
| t7 beijing / PM2.5 | Window 9 | 35064 cells · missing rate 0.02626626 · invalid 0 · nonfinite 0 · finite 34143 · 12 entities · IQR 66 |
| t7 beijing / PM2.5 | Window 10 | 35064 cells · missing rate 0.03057267 · invalid 0 · nonfinite 0 · finite 33992 · 12 entities · IQR 70 |
| t7 beijing / PM2.5 | Window 11 | 35064 cells · missing rate 0.01072325 · invalid 0 · nonfinite 0 · finite 34688 · 12 entities · IQR 139 |
| t7 beijing / PM2.5 | Comparison 0 → 1 | Support 35064/35064 · Δmissing -0.003422313 · CDF 0.1053751 · W1 13.90727 · W1/IQR 0.151166 · investigation candidate |
| t7 beijing / PM2.5 | Comparison 1 → 2 | Support 35064/35064 · Δmissing 0.001654118 · CDF 0.1267012 · W1 27.25787 · W1/IQR 0.3539984 · investigation candidate |
| t7 beijing / PM2.5 | Comparison 2 → 3 | Support 35064/35064 · Δmissing 0.005219028 · CDF 0.1161643 · W1 27.26335 · W1/IQR 0.2181068 · investigation candidate |
| t7 beijing / PM2.5 | Comparison 3 → 4 | Support 35064/35064 · Δmissing -2.851928e-05 · CDF 0.04896914 · W1 8.789447 · W1/IQR 0.1112588 |
| t7 beijing / PM2.5 | Comparison 4 → 5 | Support 35064/35064 · Δmissing 0.005789414 · CDF 0.1557137 · W1 14.34467 · W1/IQR 0.1611761 · investigation candidate |
| t7 beijing / PM2.5 | Comparison 5 → 6 | Support 35064/35064 · Δmissing -0.005817933 · CDF 0.1178911 · W1 22.47244 · W1/IQR 0.1920722 · investigation candidate |
| t7 beijing / PM2.5 | Comparison 6 → 7 | Support 35064/35064 · Δmissing 0.001682637 · CDF 0.1151705 · W1 13.71948 · W1/IQR 0.1829265 · investigation candidate |

20 of 5039 metric rows shown. 5019 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage. Long cells are abbreviated; canonical HTML/JSON retains detail. Additional charts remain in canonical HTML.

RETAINED EVIDENCE · 6 / 8

## Availability and outcome maturity

Undeclared availability remains unknown. Historical agreement concerns supplied coverage; it does not establish what a pipeline consumed.

| Evidence / source | Measurement or rule | Retained result |
|---|---|---|
| t6 beijing | features | ["PM2.5","PM10","TEMP","PRES","DEWP","RAIN","WSPM","wd"] |
| t6 beijing | meaning | Undeclared availability remains unknown; no timestamp inheritance. |

2 of 2 metric rows shown. Presentation limits do not change analytical coverage. Long cells are abbreviated; canonical HTML/JSON retains detail.

RETAINED EVIDENCE · 7 / 8

## Chronological evaluation

No retained evidence applies to this topic in the declared scope.

0 of 0 metric rows shown. Presentation limits do not change analytical coverage. Long cells are abbreviated; canonical HTML/JSON retains detail.

RETAINED EVIDENCE · 8 / 8

# Methodology and coverage

| Evidence / source | Measurement or rule | Retained result |
|---|---|---|
| method | schema | 1.4 |
| method | methodology | temporal_reliability_v1 |
| method | execution status | completed |
| method | aggregate scope | Every eligible declared record in required stages; no silent sampling. |
| method | search scope [0] / selected measurements | ["PM2.5","PM10","TEMP","PRES","DEWP","RAIN","WSPM","wd"] |
| method | search scope [0] / unsearched scope | [] |
| method | search scope [0] / work observation visits | 85836672 |
| method | search scope [0] / comparisons | 840 |
| method | search scope [0] / selection | Requested measurements first, then declaration order. Detail retention can miss behavior-only exceptions. |
| method | search scope [0] / defaults validated universal defect criteria | No |
| method | internal detail omissions / beijing:entity histories | 0 |
| method | internal detail omissions / beijing:wd:window categories | 132 |
| method | internal detail omissions / beijing:wd:window categories:… [JSON] | 132 |
| method | internal detail omissions / beijing:series effects | 236 |
| method | timezone / package | tzdata |
| method | timezone / version | 2026.5 |
| method | timezone / zones / Asia/Shanghai | bf8b7ed82fe6e63e6d98f8cea934eeac901cd16aba85eb5755ce3f8b4289ea8a |
| method | phase seconds / beijing:ingestion | 3.361883 |
| method | phase seconds / beijing:normalization | 5.319458 |
| method | phase seconds / required diagnostics | 3.720628 |

20 of 36 metric rows shown. 16 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage. Long cells are abbreviated; canonical HTML/JSON retains detail.

Decoded-batch controls bound accepted batches and application retention. Native pages, dictionaries and conversion buffers can allocate before checks. Estimates and workload limits are provisional engineering controls. RSS monitoring can overshoot between samples.

Analytical scope left unsearched, retained-detail omissions, and presentation omissions are separate. No independent-row significance testing, model fitting, split generation, automatic repair or universal temporal score is performed.

# Retained witnesses and selected detail

100 of 236 retained detail rows available here. Print shows up to 20 rows; JSON preserves every retained record. These presentation omissions do not change searched scope.

| Evidence | Detail | Witness / selected population |
|---|---|---|
| t4 | retained histories | {"entity": ["Aotizhongxin"],"measurement":"DEWP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000… [JSON] |
| t4 | retained histories | {"entity": ["Aotizhongxin"],"measurement":"PM10","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000… [JSON] |
| t4 | retained histories | {"entity": ["Aotizhongxin"],"measurement":"PM2.5","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"1262268000… [JSON] |
| t4 | retained histories | {"entity": ["Aotizhongxin"],"measurement":"PRES","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000… [JSON] |
| t4 | retained histories | {"entity": ["Aotizhongxin"],"measurement":"RAIN","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000… [JSON] |
| t4 | retained histories | {"entity": ["Aotizhongxin"],"measurement":"TEMP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000… [JSON] |
| t4 | retained histories | {"entity": ["Aotizhongxin"],"measurement":"WSPM","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000… [JSON] |
| t4 | retained histories | {"entity": ["Aotizhongxin"],"measurement":"wd","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"1262268000000… [JSON] |
| t4 | retained histories | {"entity": ["Changping"],"measurement":"DEWP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000000… [JSON] |
| t4 | retained histories | {"entity": ["Changping"],"measurement":"PM10","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000000… [JSON] |
| t4 | retained histories | {"entity": ["Changping"],"measurement":"PM2.5","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"1262268000000… [JSON] |
| t4 | retained histories | {"entity": ["Changping"],"measurement":"PRES","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000000… [JSON] |
| t4 | retained histories | {"entity": ["Changping"],"measurement":"RAIN","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000000… [JSON] |
| t4 | retained histories | {"entity": ["Changping"],"measurement":"TEMP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000000… [JSON] |
| t4 | retained histories | {"entity": ["Changping"],"measurement":"WSPM","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000000… [JSON] |
| t4 | retained histories | {"entity": ["Changping"],"measurement":"wd","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"1262268000000000… [JSON] |
| t4 | retained histories | {"entity": ["Dingling"],"measurement":"DEWP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000… [JSON] |
| t4 | retained histories | {"entity": ["Dingling"],"measurement":"PM10","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000… [JSON] |
| t4 | retained histories | {"entity": ["Dingling"],"measurement":"PM2.5","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"12622680000000… [JSON] |
| t4 | retained histories | {"entity": ["Dingling"],"measurement":"PRES","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000… [JSON] |